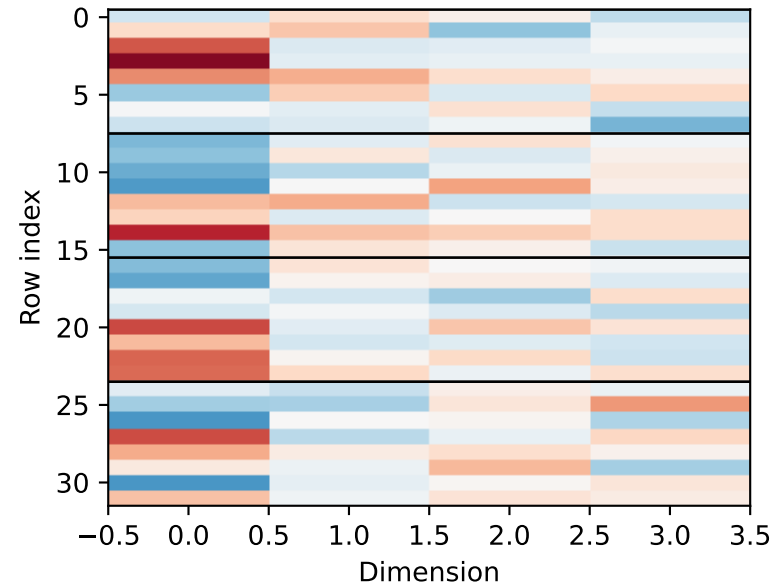
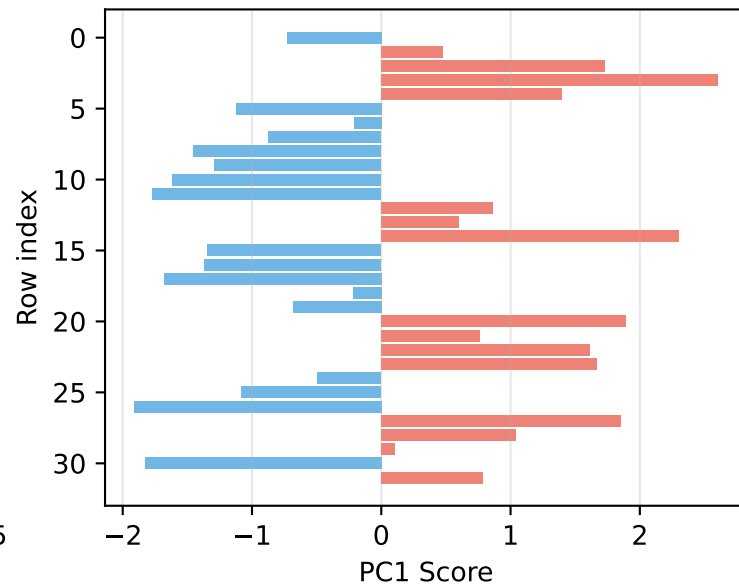


PCA Sort Pipeline: $O(N \log N)$ Sort $\rightarrow$ Better Block-Mean Approximation

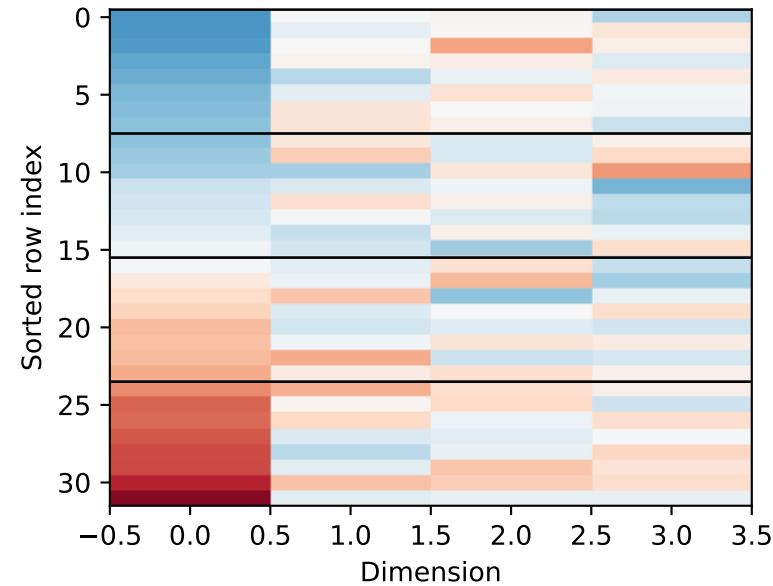
**Step 1: Cold Rows
(original order)**



**Step 2: Compute PC1
(1 PCA on cold rows)**



**Step 3: Sort by PC1
(similar rows adjacent)**



**Step 4: Block Mean
(each block $\rightarrow$ 1 value)**

